

Goal

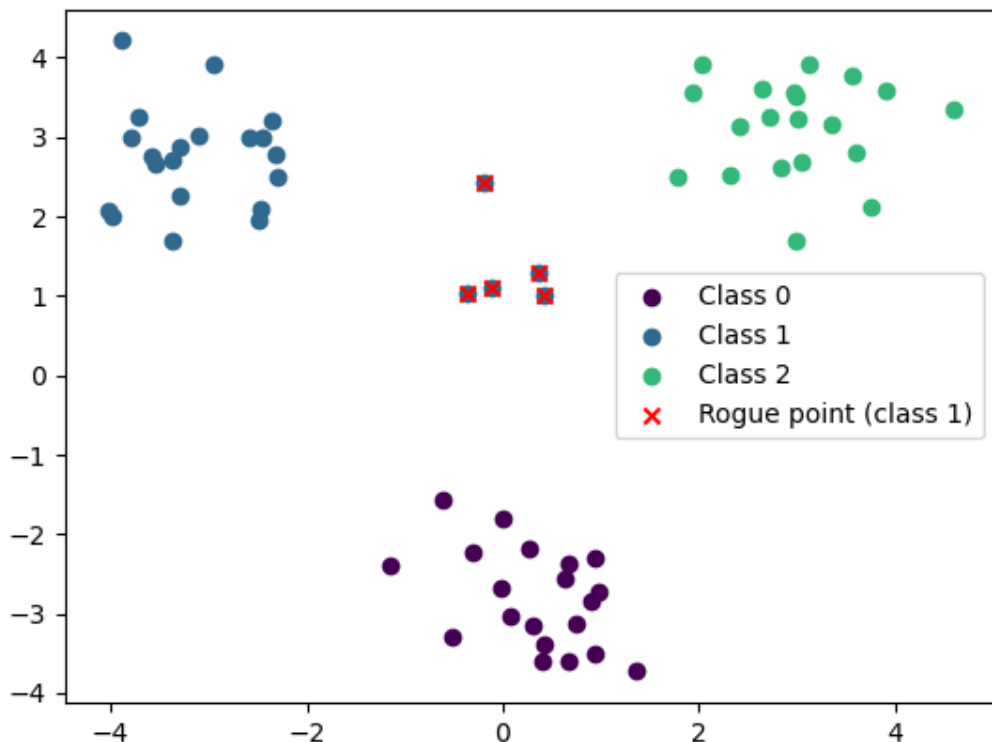
In this experiment we want to investigate the behavior of the SSD method and how the choice of hyperparameters affect the decision boundary. We also try to address the shortcomings of SSD in two ways:

- 1. By applying dampening at only the last layer of the model. This is referred to as "SSD v2"
- 2. By searching for hyperparameters whose dampening yields a loss as close as possible to the original model's loss on a "generalization dataset" constructed in the same way as the repair step for Scrub+R. This is referred to as "SSD v3".
- 3. A combination of the previous two approaches. This is referred to as "SSD v4".
- 4. Here we learn two sets of parameters: (α_1, λ_1) which are applied to the first half of the network and (α_2, λ_2) which are applied to the second half. Additionally, when constructing the generalization set we check if all classes are present in forget. If not, additional points for the missing classes are added BOTH to the forget set and the generalization set. We hypothesize that this will favor parameters that do not induce catastrophic forgetting.

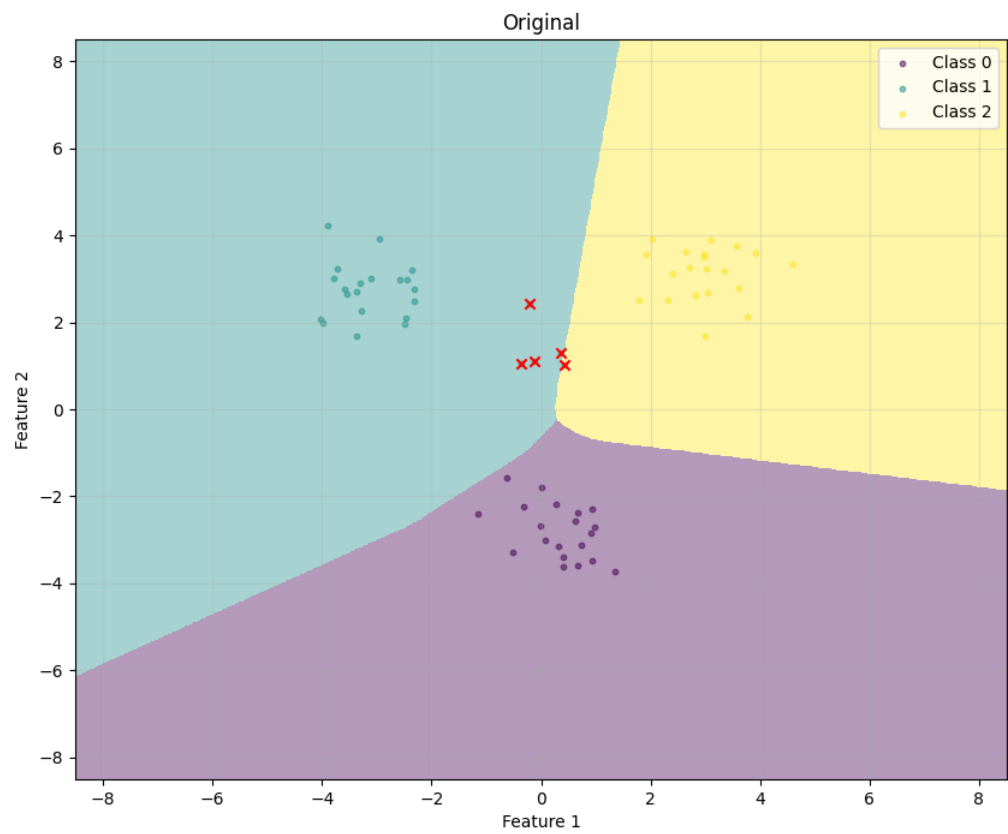
Experiment runs

Now we run the experiment on each of the dataset configuration, moving the rogue point.

1. Rogue point with same distance to all centroids (close to decision boundary)



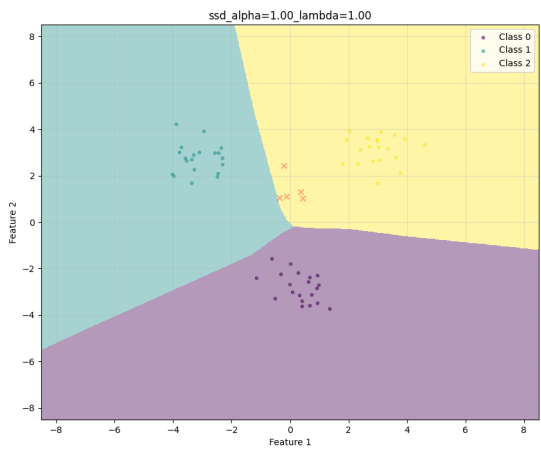
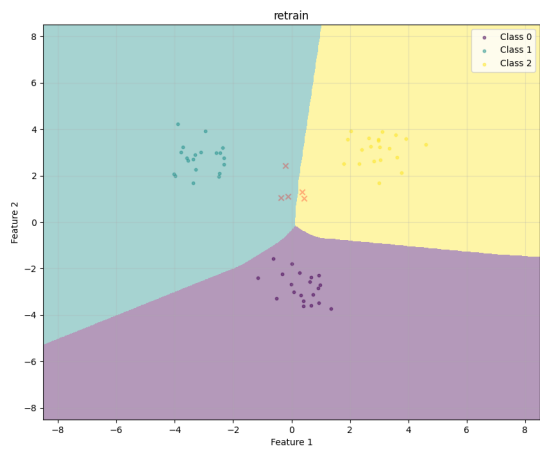
Original model for the dataset

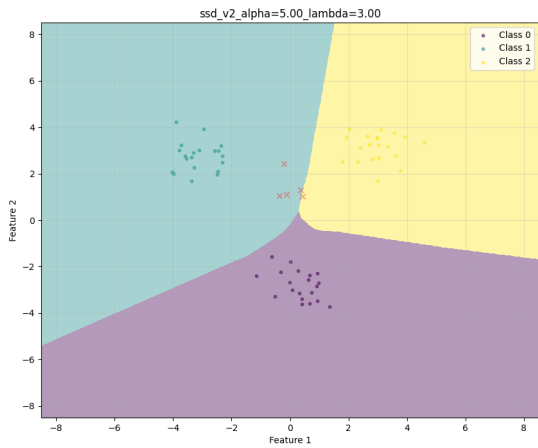
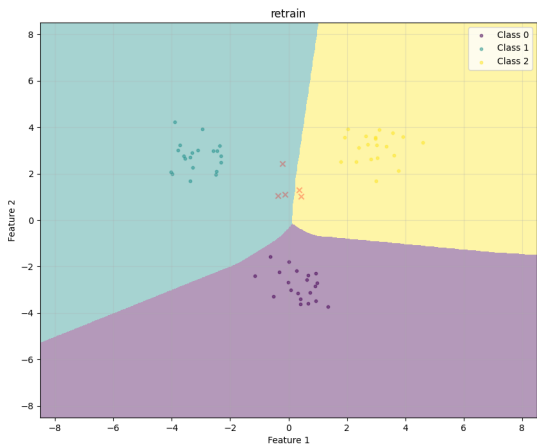
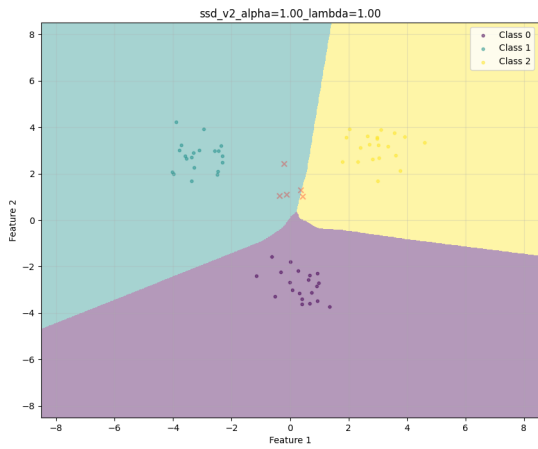
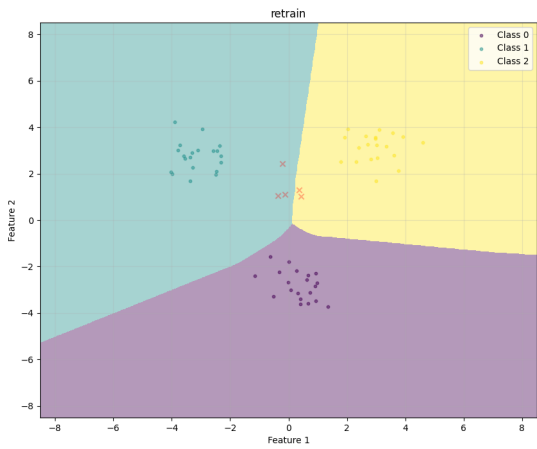
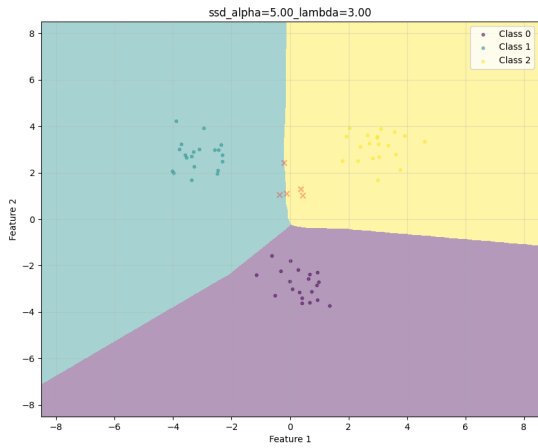
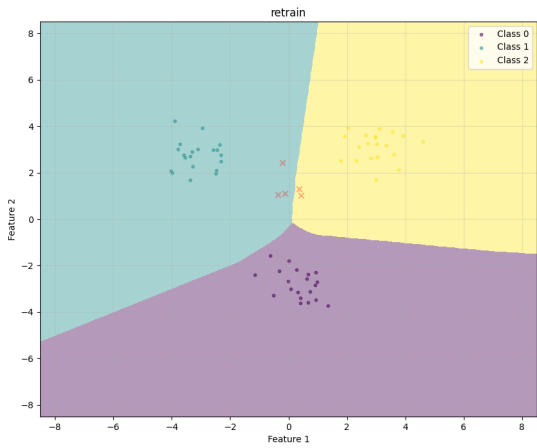


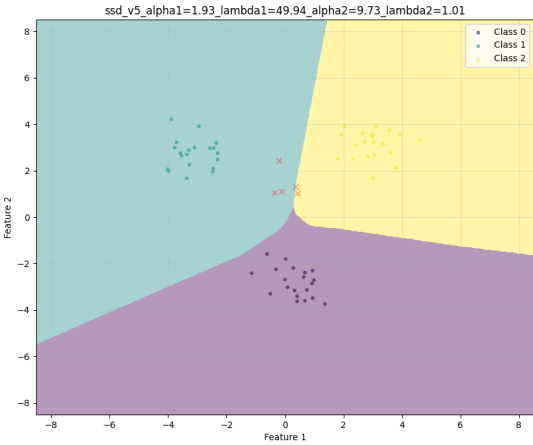
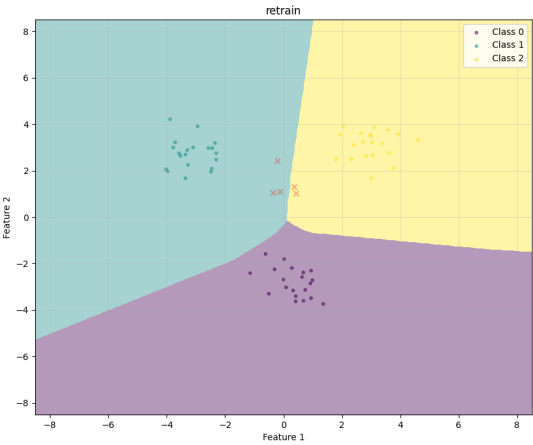
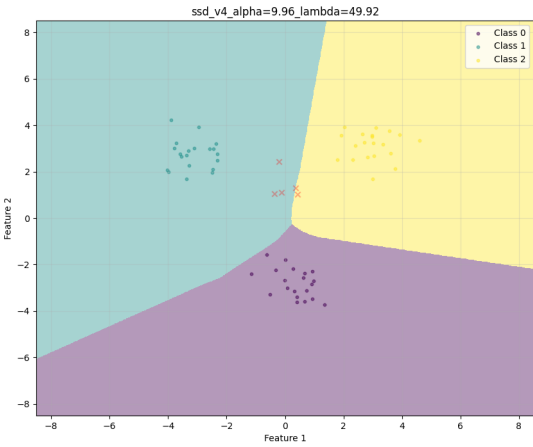
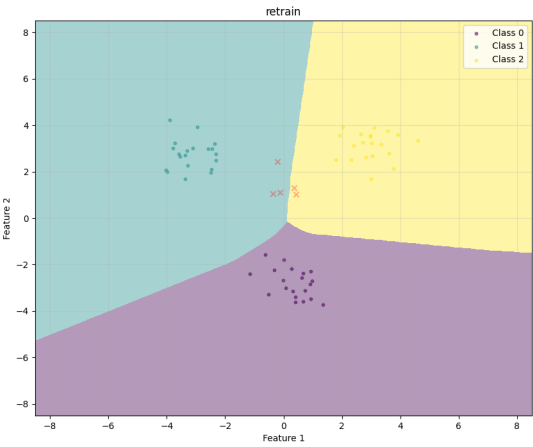
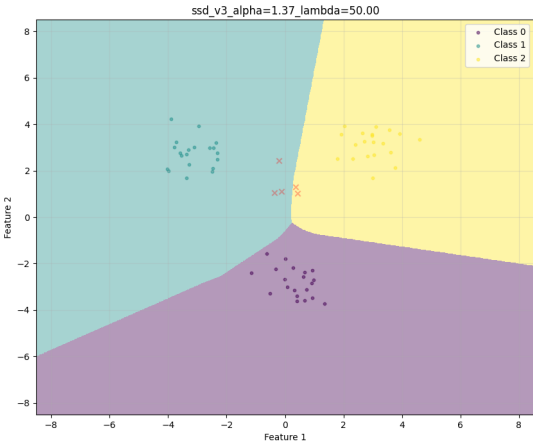
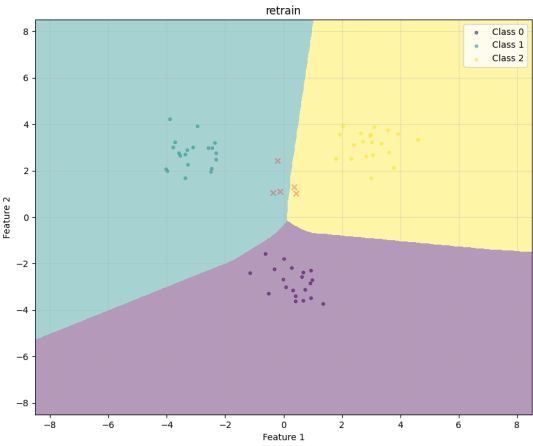
Findings

Unlearned

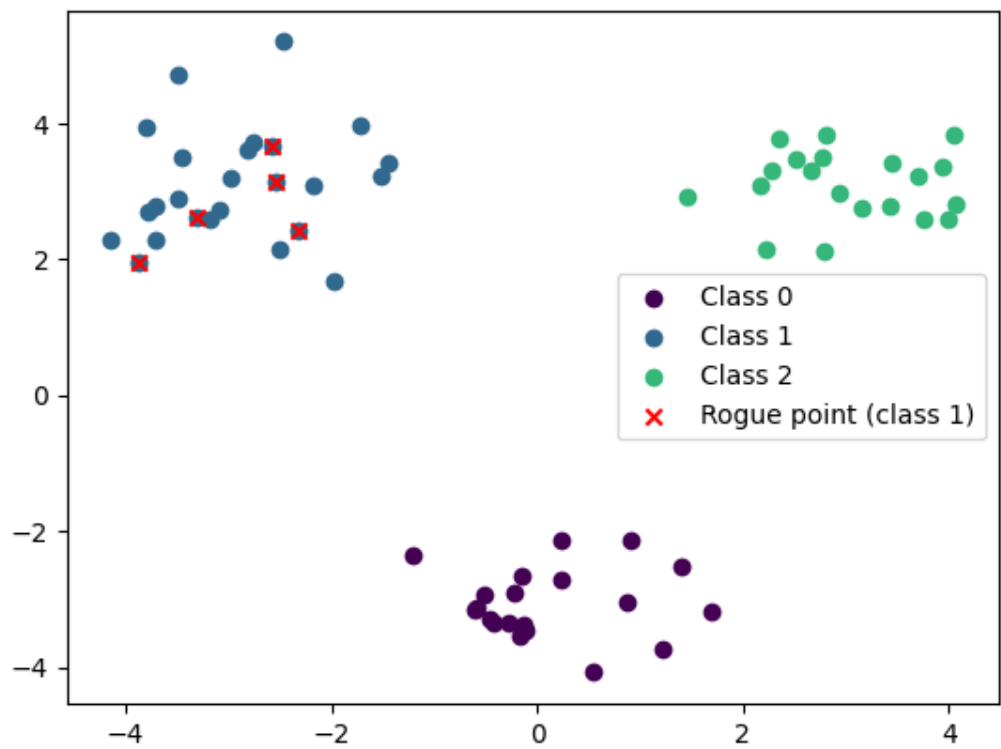
SSD



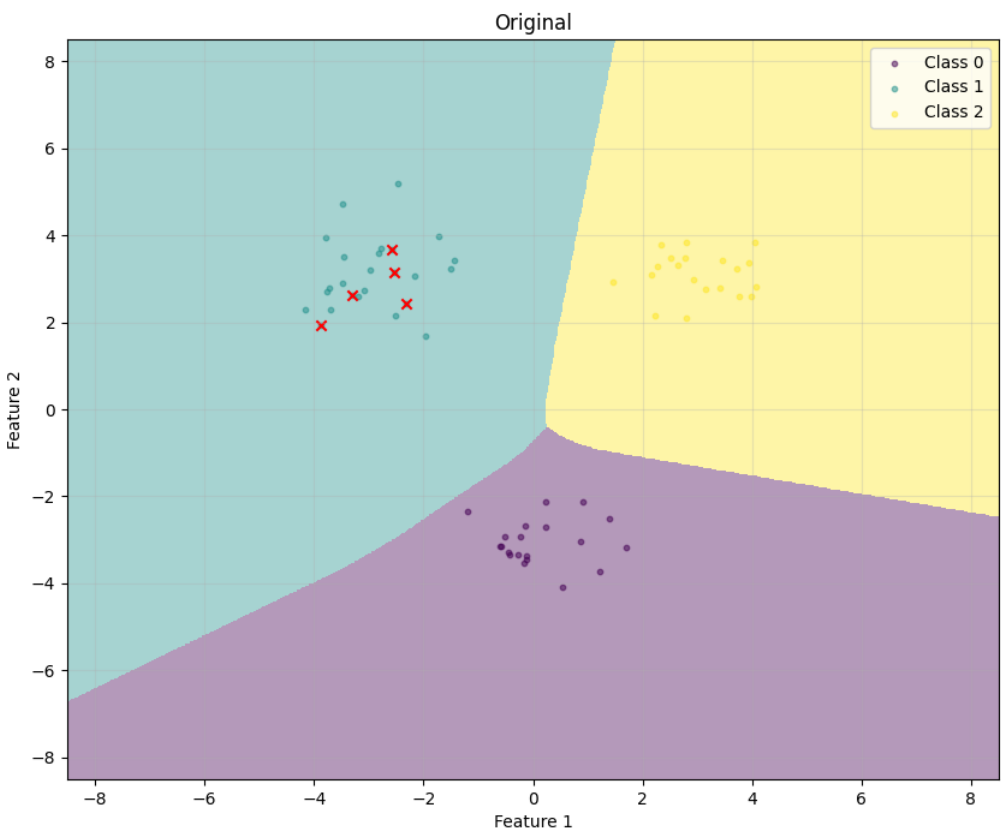




2. Rogue point with same centroid as its class



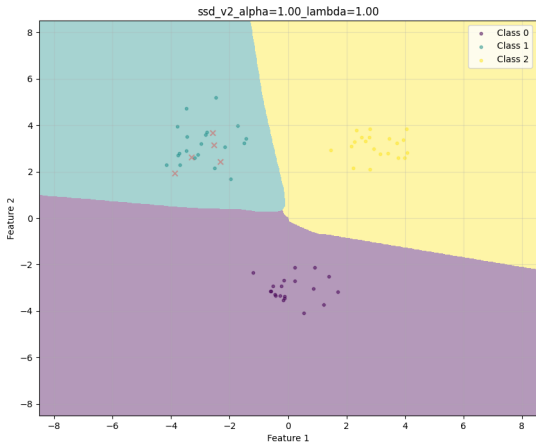
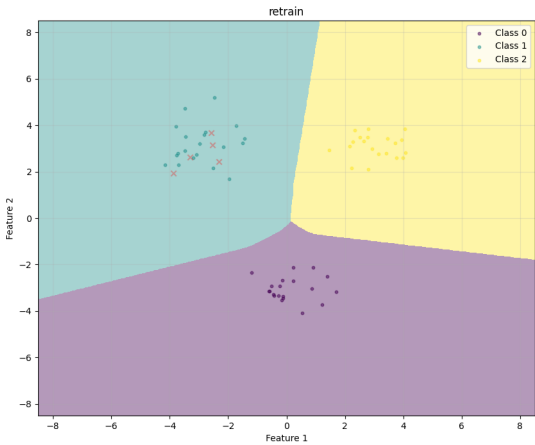
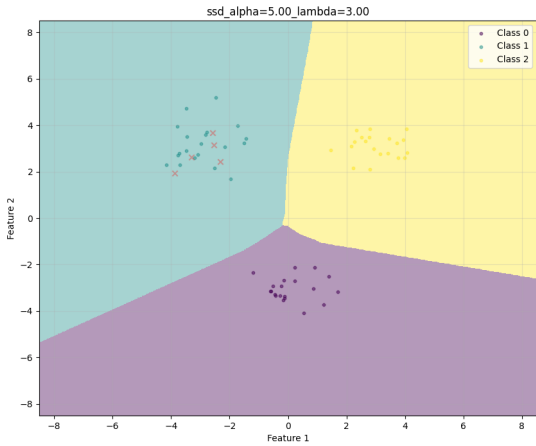
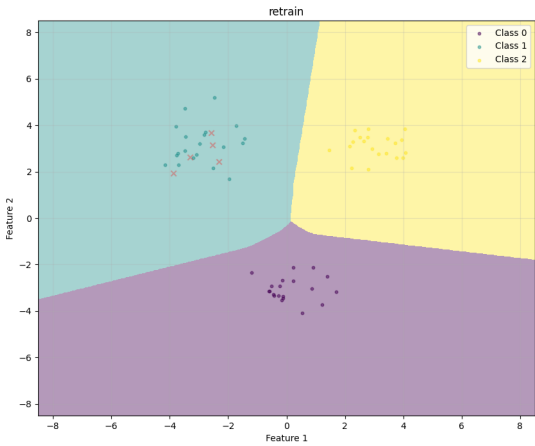
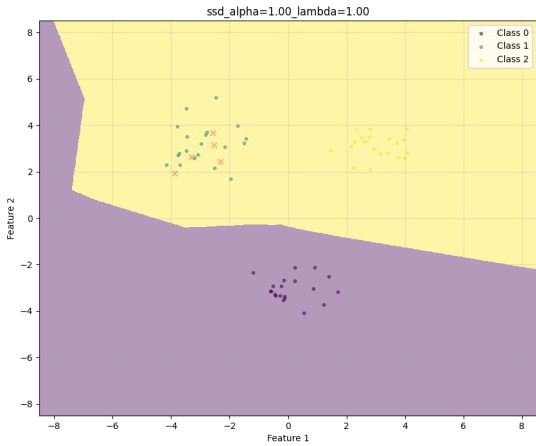
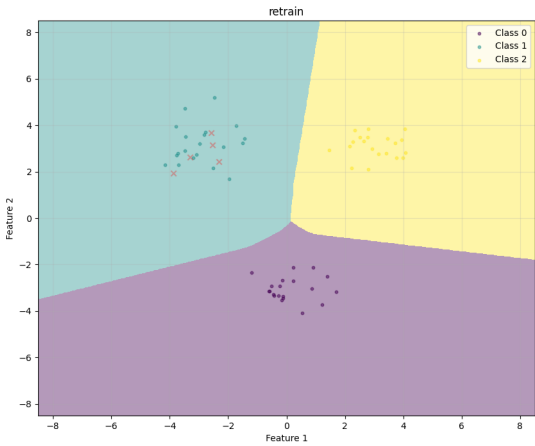
Original model for the dataset

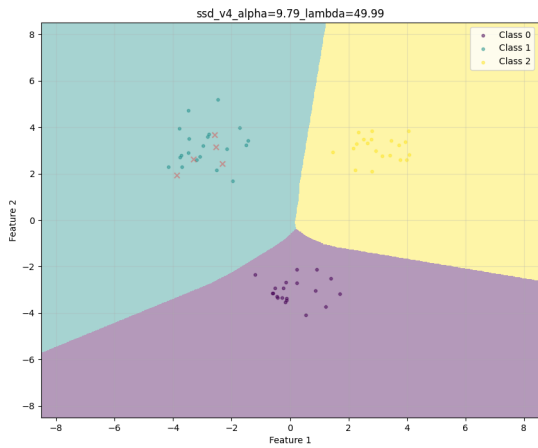
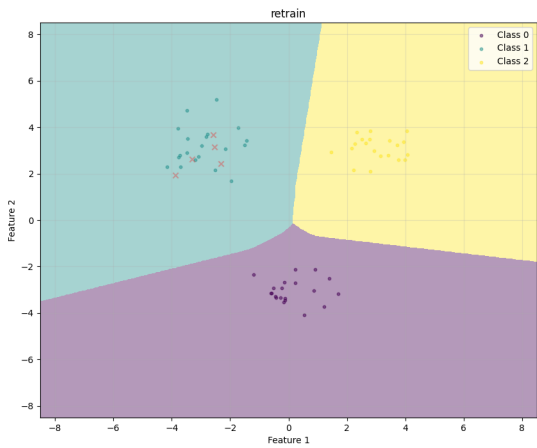
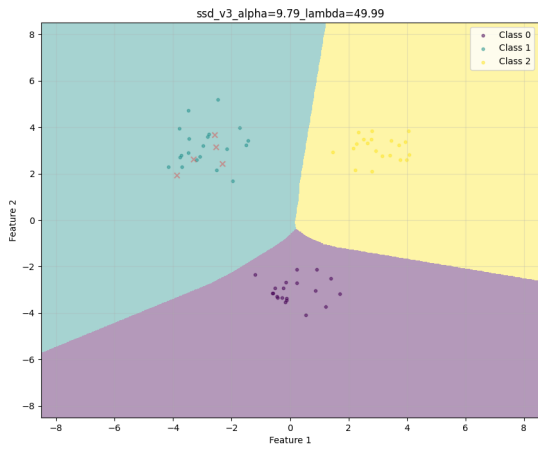
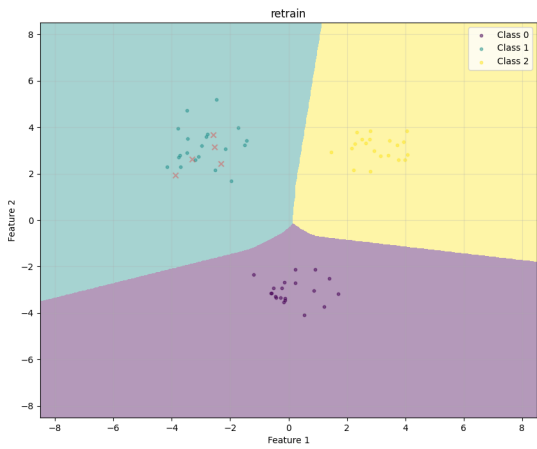
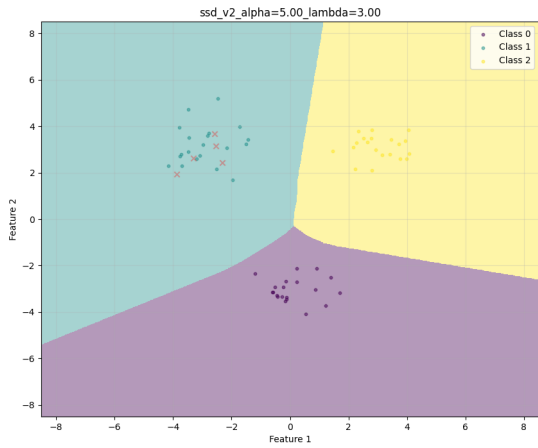
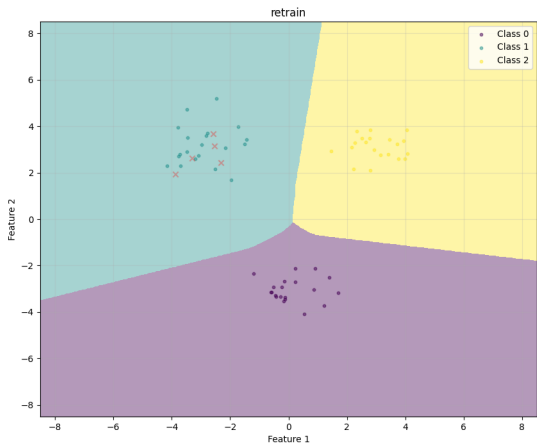


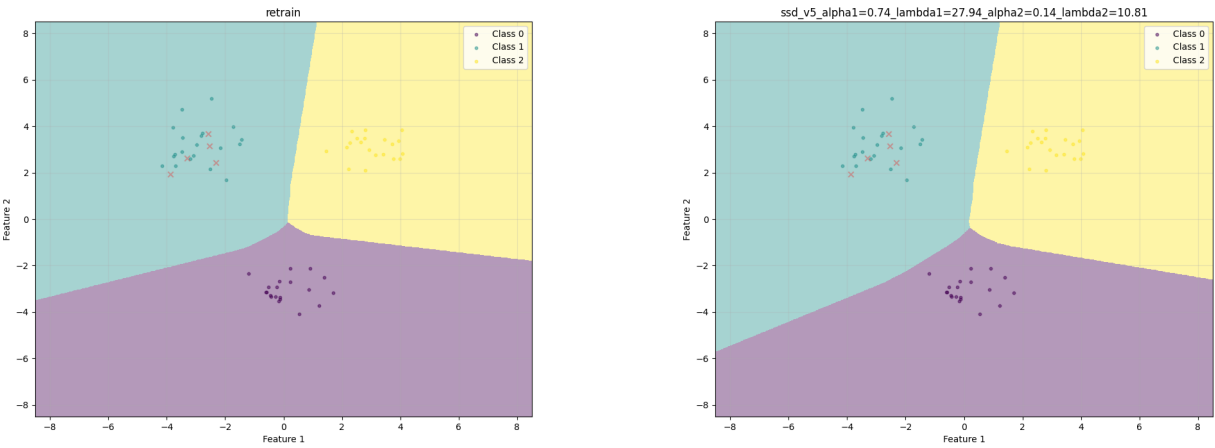
Findings

Unlearned

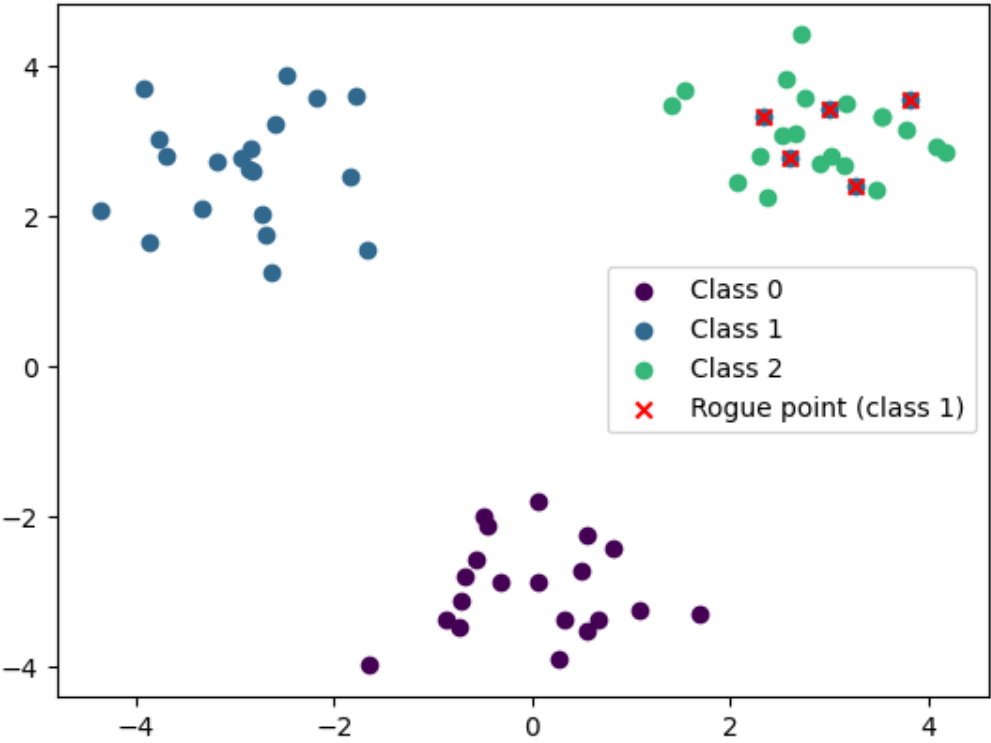
SSD



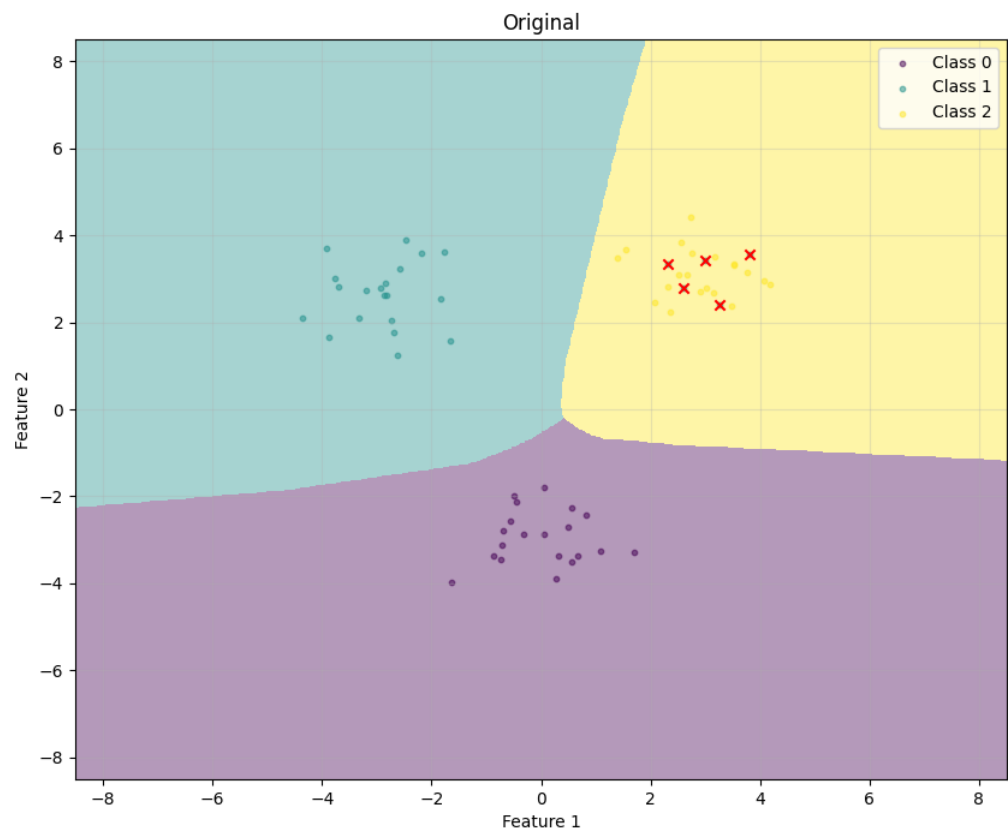




3. Rogue point with same centroid as the class with a different label



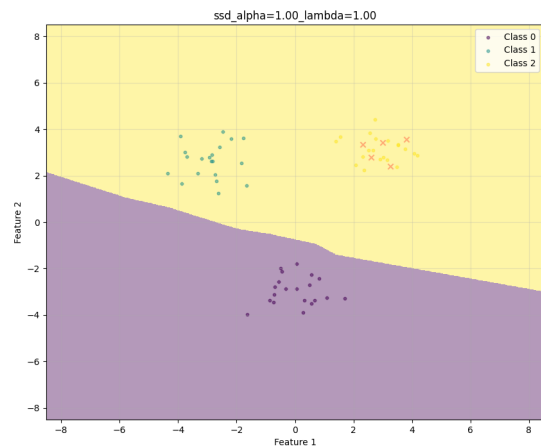
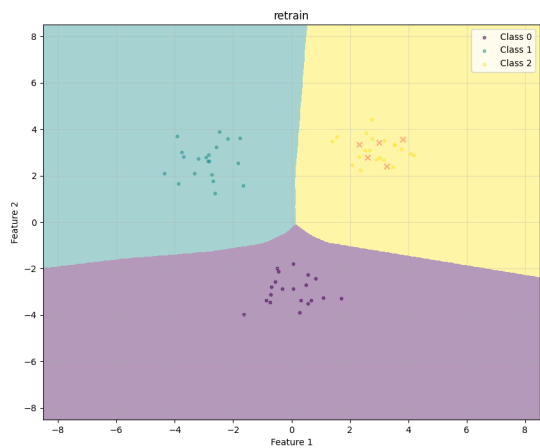
Original model for the dataset

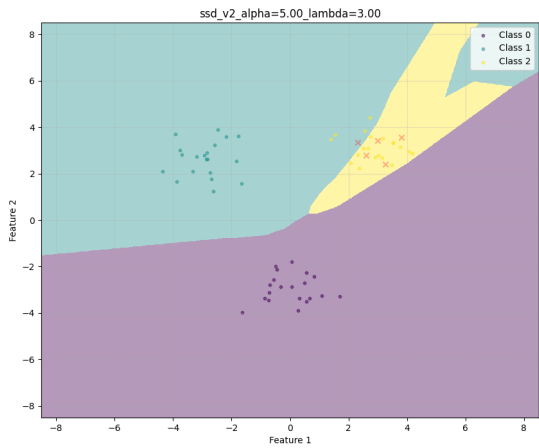
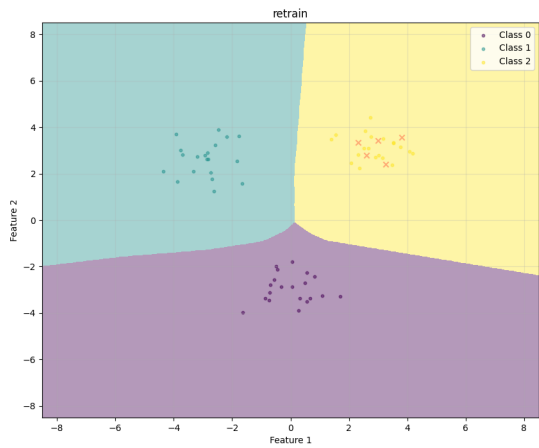
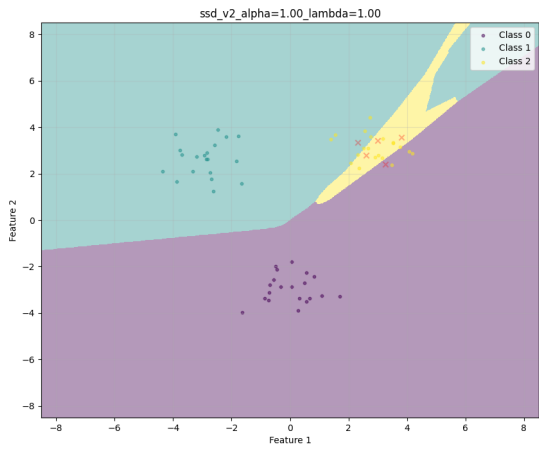
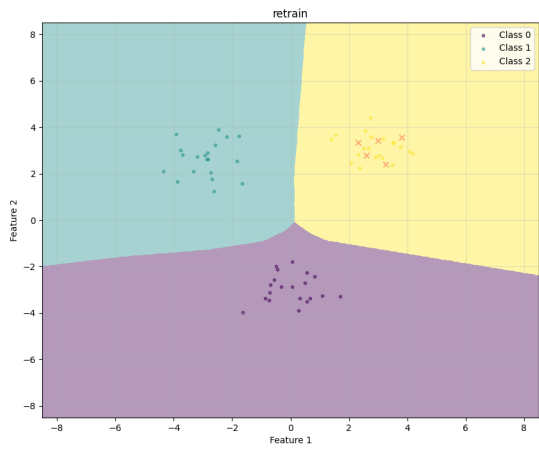
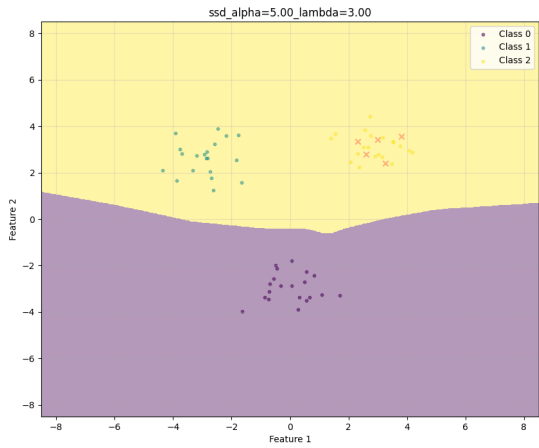
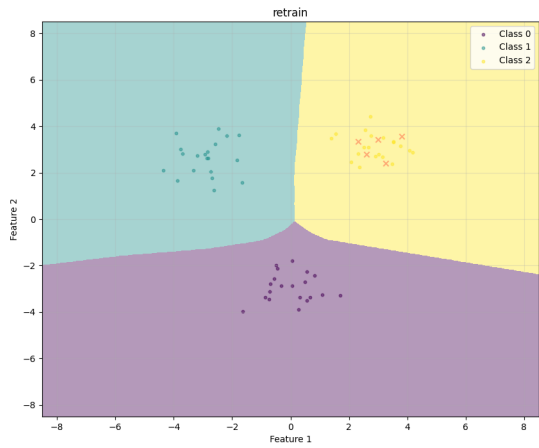


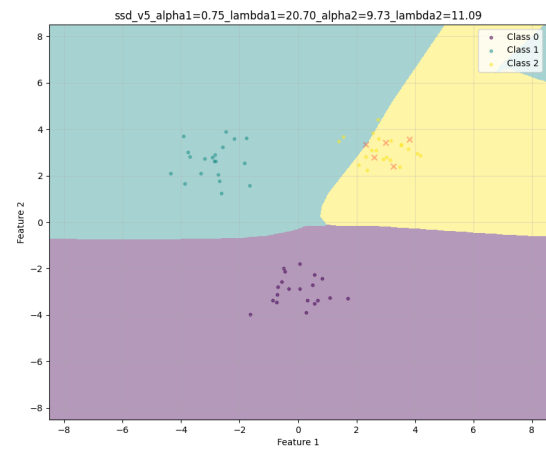
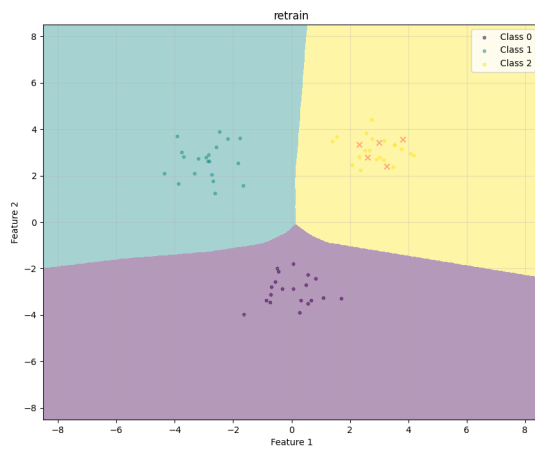
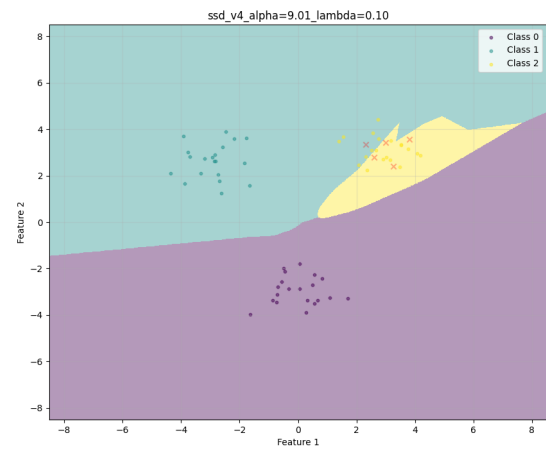
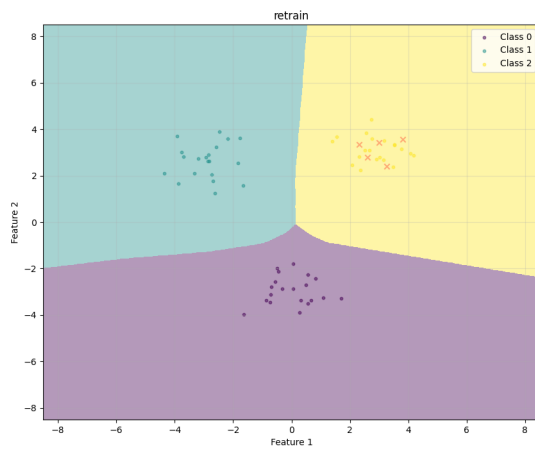
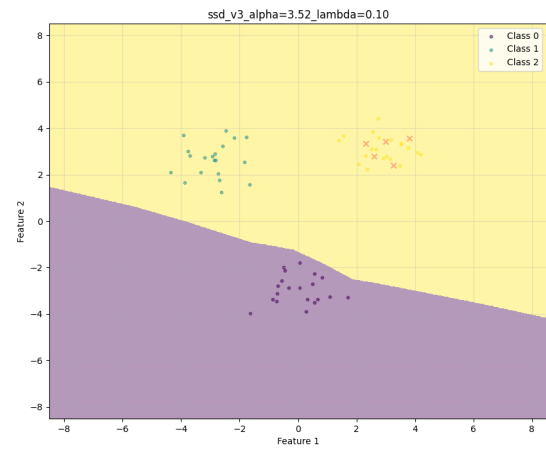
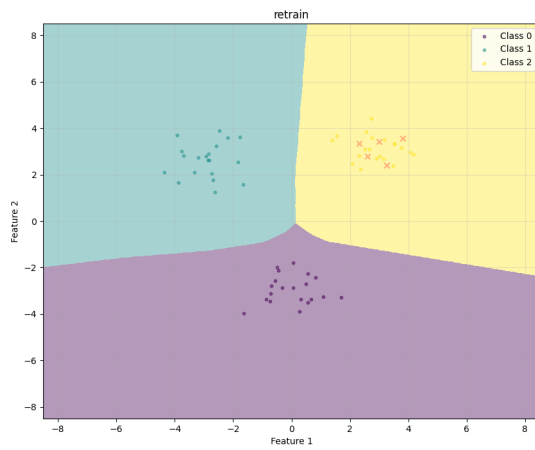
Findings

Unlearned

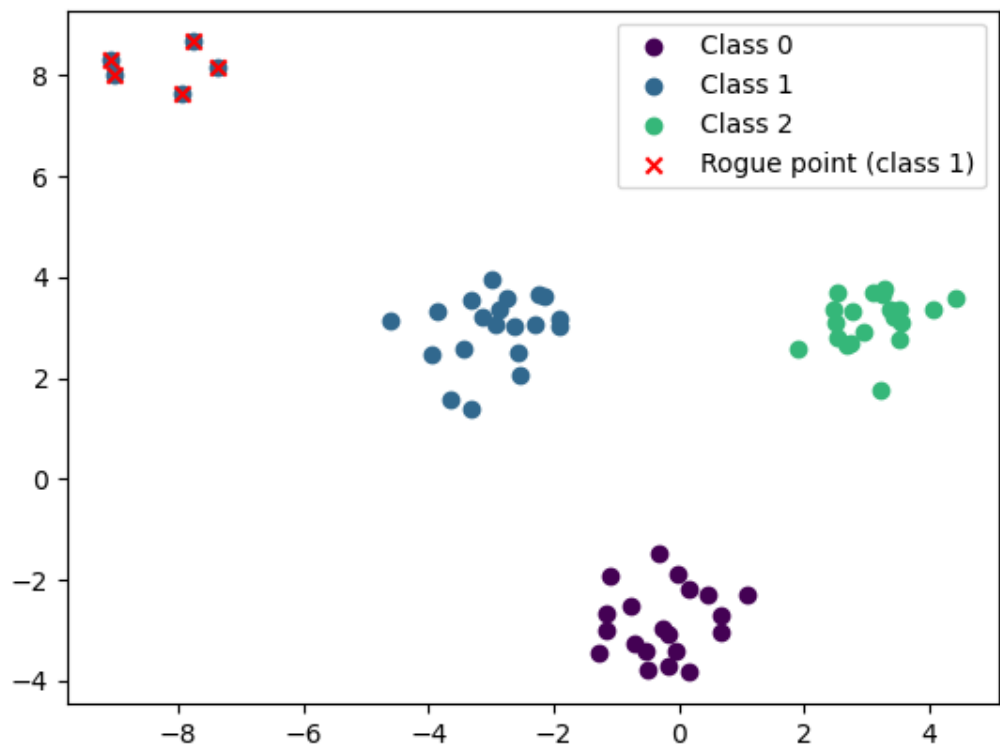
SSD



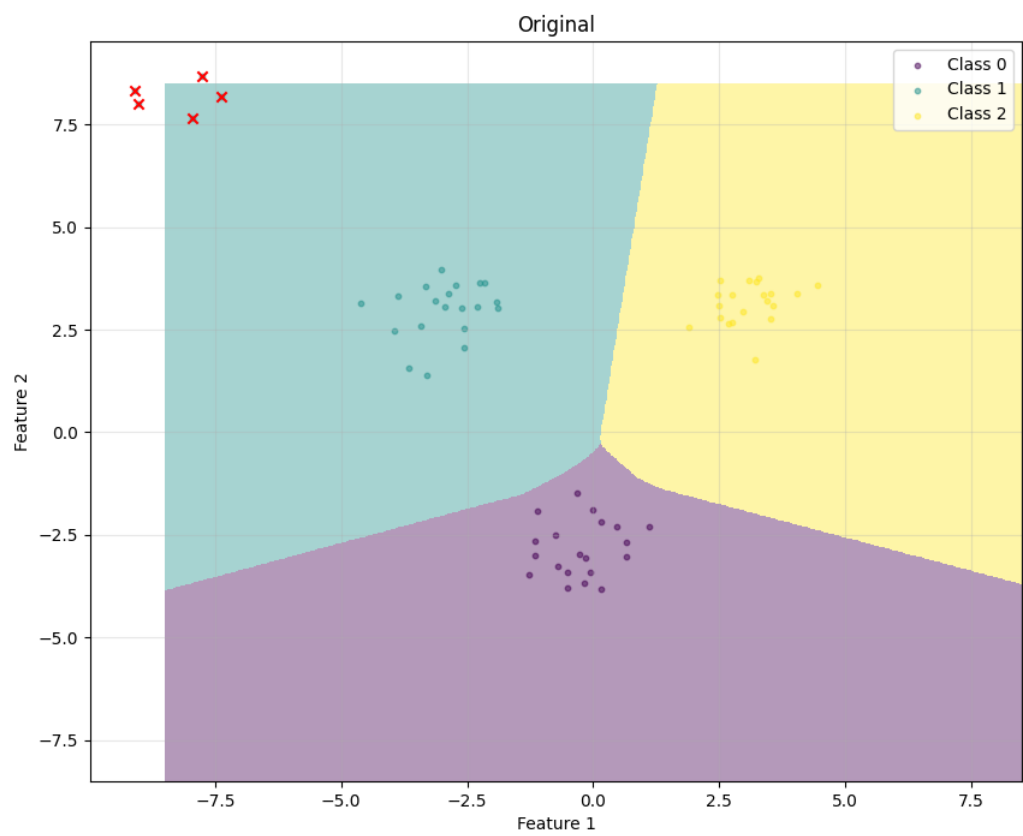




4. Rogue point far away from its centroid, but probably in the same decision boundary



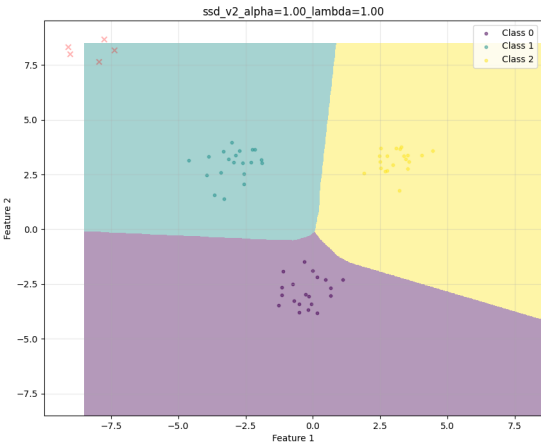
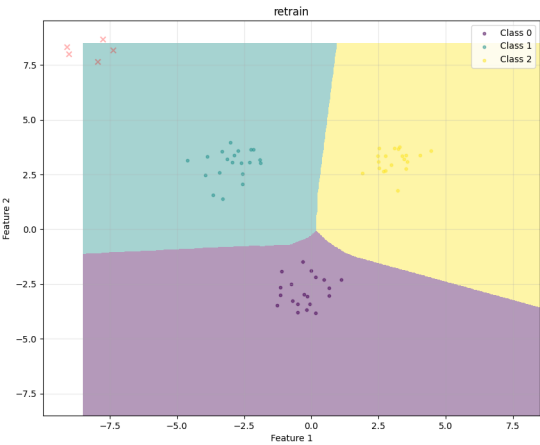
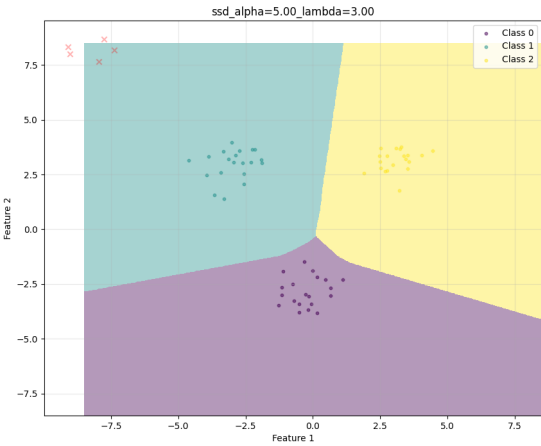
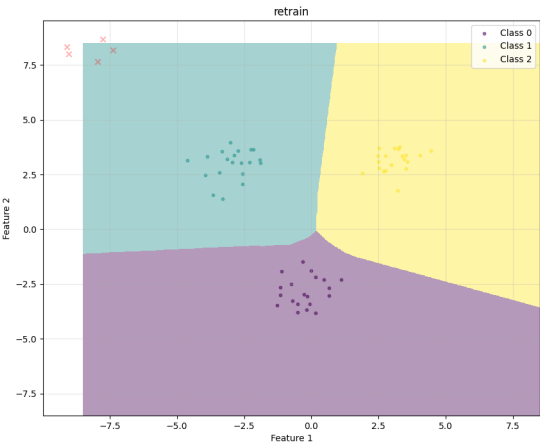
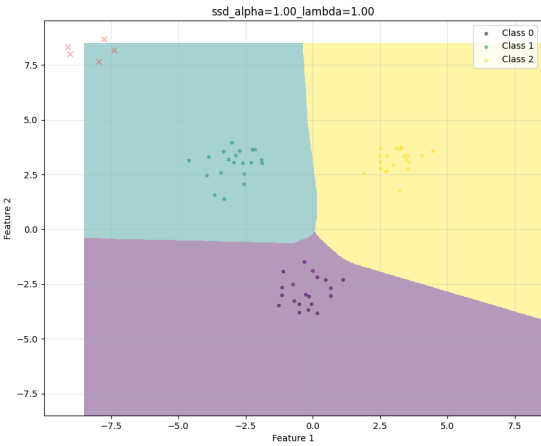
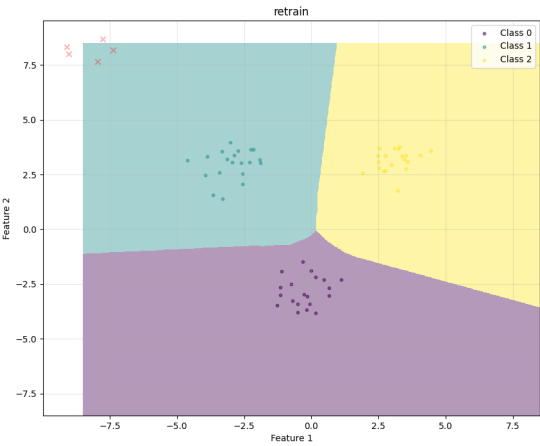
Original model for the dataset

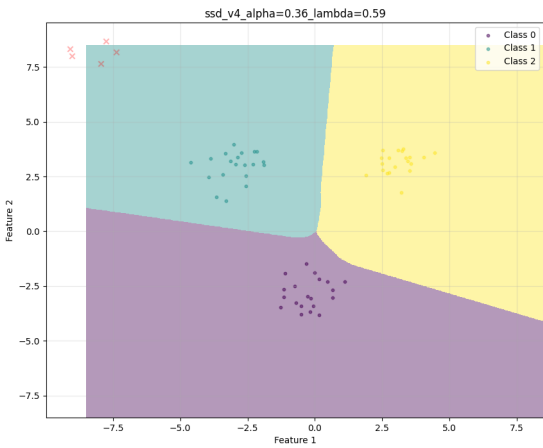
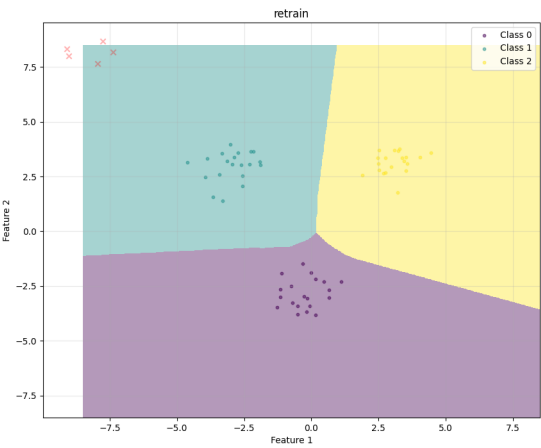
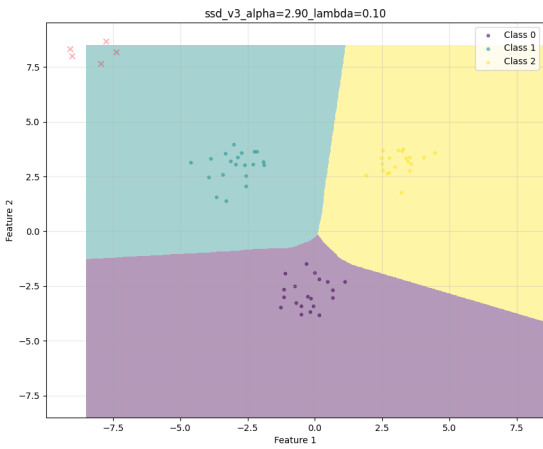
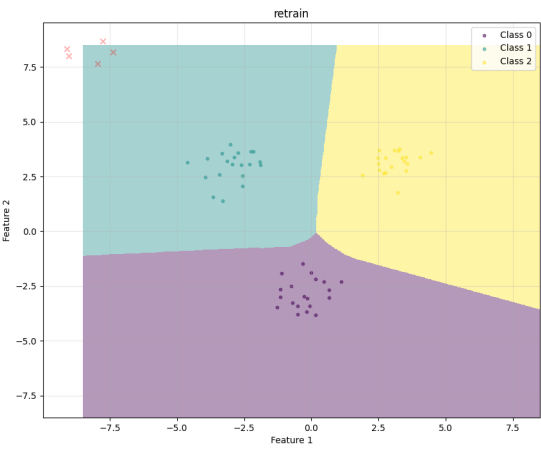
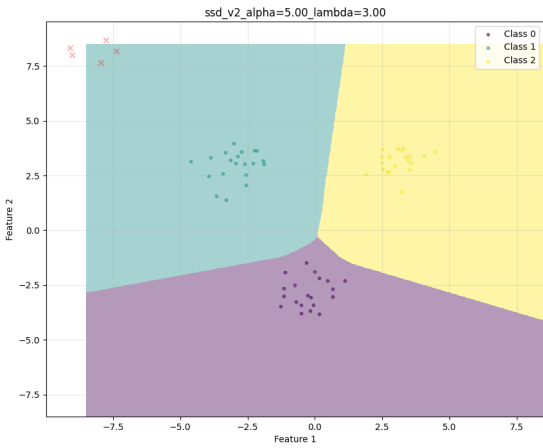
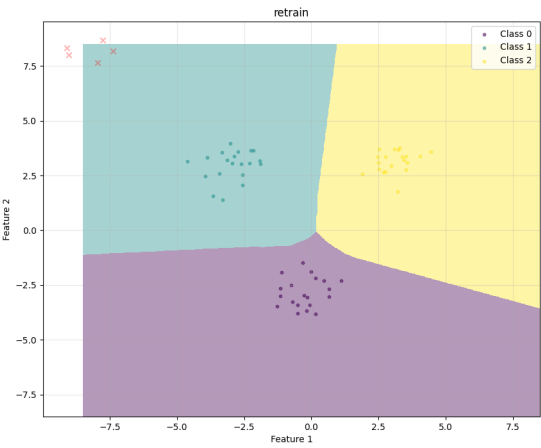


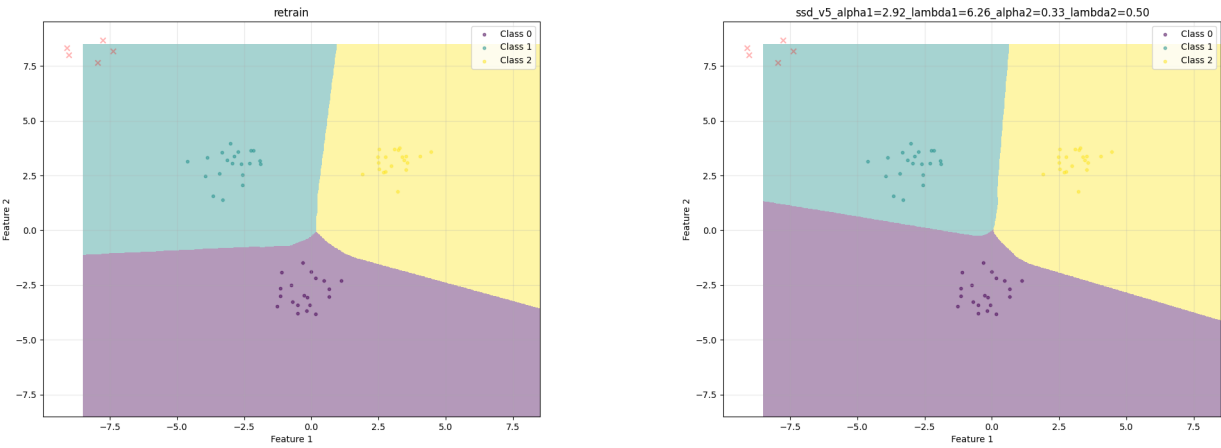
Findings

Unlearned

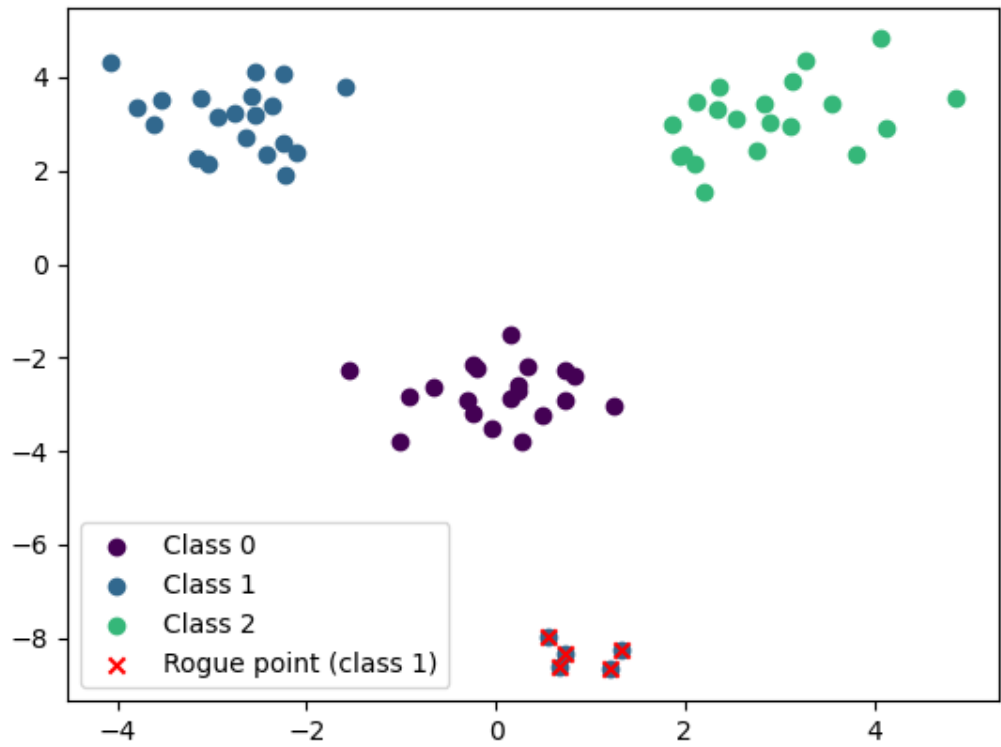
SSD



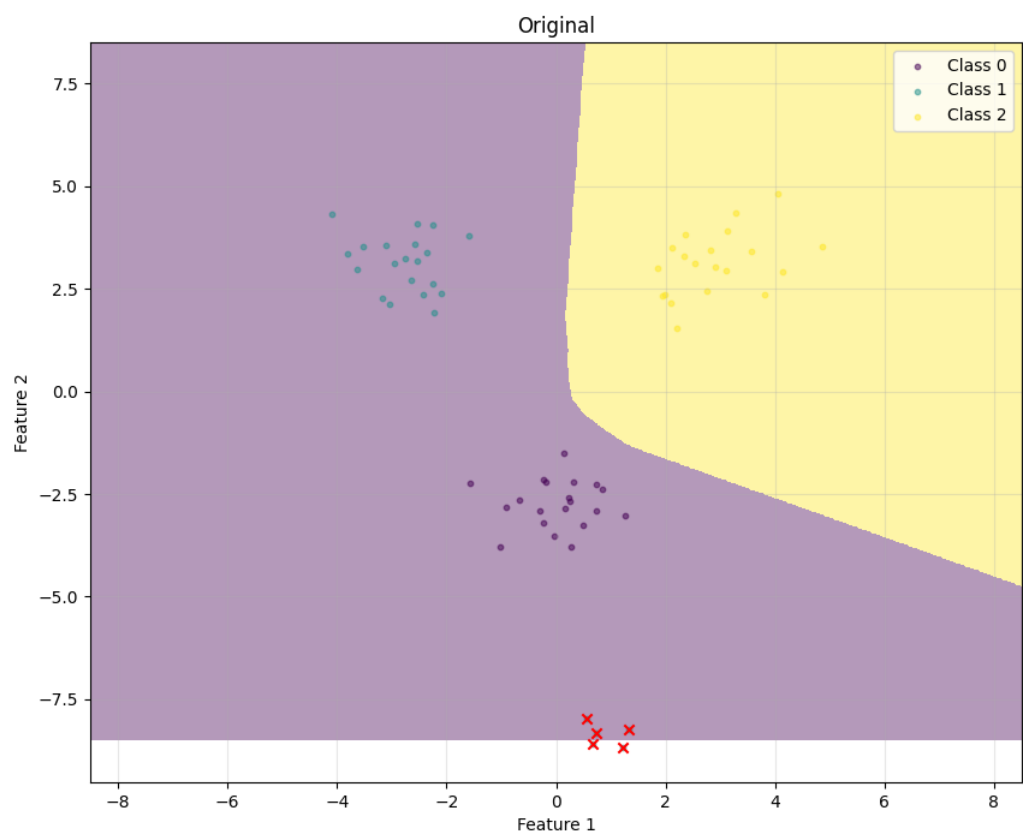




5. Rogue point far away from its centroid, but probably in the same decision boundary



Original model for the dataset



Findings

Unlearned

SSD

